Lars Larsen

Electronics Design Engineer: analog, mixed-signal and video hardware

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SUMMARY

Electronics engineer with 15 years of analog, mixed-signal and video design. Led design work on more than 100 LZX Industries products with a three-person core team, including discrete video-rate analog circuits (Gilbert-cell and matched-pair multipliers, keyers, sync), video I/O from composite to HDMI, power entry boards, and the FPGA and microcontroller systems that control them.

EXPERIENCE

Founder & Principal Electronics Engineer, LZX Industries

Portland, OR · 2011 – Present

Maker of video synthesizers and video processing instruments (DBA 2011, LLC 2017). Led product design in a three-person core team, working with partners who handle firmware/HDL and board layout.

- Led product design as our team took more than 100 hardware products to production across six Eurorack series, standalone instruments, DIY kits, cases and power. About 240 production circuit board designs and 140 metal front panels. $6.4M in direct sales (2018 to 2026), plus a dealer channel.
- Videomancer (shipping since late 2025): designed a standalone FPGA video synthesizer around an iCE40 HX4K and RP2040, with ADV7611/ADV7513 HDMI in and out, ADV7181C and ADA4412-3 analog input, and an ADV7393 SD/ED/HD encoder. Users load effects as Ed25519-signed packages. Released 69 public firmware release candidates and an open GPL-3.0 SDK.
- Chromagnon (RevJ prototype on the bench): led a video-rate analog/digital instrument through seven board revisions to an 8-layer core. Replaced the Zynq SoM with an iCE40 and RP2040 controlling an analog Gilbert-cell XY engine, after costing an all-digital dual-ECP5 version. On-board silicon cost fell 64% ($195 to $70) and BOM cost a further 26%.
- Designed analog video circuits for shipped product lines: matched-pair multipliers and blenders, keyers, colorspace mapping, sync separation and video-rate op-amp stages. Co-defined the LZX 1V signal convention (output stages, 100 kΩ inputs) used in every Eurorack series.
- Gen3 power entry boards (FPGA12, SYNC12, PSU8, PSU16; shipped, revised 2021 to 2023): one +12 V input with diode-OR and EMI filter, a synchronous buck and an inverting converter to ±5.5 V, and RF LDOs down to clean ±5 V for analog video. Sequenced iCE40 rails and separately filtered clock and sync supplies. Used in about 15 Eurorack modules and TBC2.
- Discrete high-speed analog: Chromagnon's XY engine applies an affine transform to video-rate scan ramps using MC1496 Gilbert-cell rotators, discrete matched-pair two-quadrant multipliers and VCAs, a diode keyer and ±5 V 130 MHz op-amp stages. Reduced it from 28 multiplier cells with 56 isolated supply rails to 8, removing 190 placements.
- Simulation (2026, AI-assisted tools): ngspice reviews with derating rules and IBIS-based driver models. A 200-check review of a clock board found 9 problems, fixed before fabrication. Shared model library of about 60 parts.
- Mechanical: sheet-metal enclosures (Videomancer, made by HLH Prototypes; Eurorack cases and bus shields from SendCutSend), injection-molded custom knobs and about 140 metal front panels, modeled in Inventor and FreeCAD.
- Manufacturing: built the line through about 40 active suppliers (JLCPCB, HLH Prototypes, SendCutSend, DigiKey). Ran 1,281 manufacturing orders in MRPeasy, moved to InvenTree and ERPNext in 2026. Directed board layout by a partner.

Operations Manager, Darkplace Manufacturing

Portland, OR · 2015 – 2017

Manufacturing arm for the LZX Industries product line.

- Managed operations, purchasing and customer communication for a manufacturing team of 10+ people building the LZX product line, while continuing as its product designer.

Earlier

- Multimedia Programmer, University of North Texas, Denton TX, 2008–2010 and 2011–2015
- Lead Service Technician, Switched On (synthesizer sales and repair), Austin TX, 2010

- New Media Developer, FUNimation Entertainment, Flower Mound TX, 2007–2008
- Multimedia Developer, TargetScope, Plano TX, 2006–2007

SELECTED PROJECTS

- **Chromagnon web simulator**: Browser version of the Chromagnon instrument model (WebGL), checked against reference vectors. lzxindustries.net/instruments/chromagnon/tools/simulator
- **LZX documentation**: Manuals, patch guides and specifications for the product line (open source, github.com/lzxindustries/lzxdocs). docs.lzxindustries.net
- **Cadet open hardware**: DIY video synth module kits, open-sourced in 2022 with schematics, layouts and BOMs. github.com/lzxindustries/lzxcadet

SKILLS

Electronics	Analog, mixed-signal and video (SD to 1080p, composite, S-Video, YPbPr, RGB, HDMI, genlock), discrete high-speed analog, power entry and regulation, ngspice/SPICE, IBIS, DipTrace, Eagle, KiCad, DFM, BOM costing, board bring-up, oscilloscope automation
Firmware & FPGA	C/C++ (C++17/20, no heap), Zynq-7000 bare-metal and FreeRTOS, STM32F4, RP2040/RP2350, AVR, TinyUSB, Ed25519 signing, VHDL, Verilog, GHDL, VUnit, Yosys/nextpnr (iCE40), Vivado (Zynq-7000)
Mechanical	Inventor, FreeCAD, build123d, sheet-metal enclosures, injection-molded knobs, front panel graphics, Blender rendering
Product & operations	Concept to production, PLM (MRPeasy, InvenTree), ERP, supplier and contract manufacturer management, pricing, documentation, open-source hardware
Software	Python (NumPy, OpenCV, PySide6), TypeScript/React (Shopify Hydrogen/Remix, Vitest, Playwright), Docker, Forgejo CI, Linux servers, AI coding tools (Claude Code, Cursor)

EDUCATION

- B.A., Radio, Television & Film, University of North Texas, 2006